

Subject

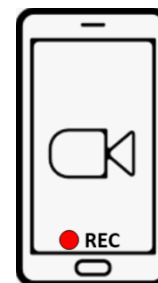
Speech and Natural Language Processing

Professor

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Participar



Session 06

Unit I

Natural Language Processing

Topic 1.7

Evaluation metrics: Accuracy, precision, recall, F1 score.

Let's review:
S06_3_Exercises_before_solution

Text Classification and Naive Bayes

Precision, Recall, and F1

Evaluating Classifiers: How well does our classifier work?

Let's first address binary classifiers:

—Is this email spam?

spam (+) or not spam (-)

—Is this post about Delicious Pie Company?

about Del. Pie Co (+) or not about Del. Pie Co(-)

We'll need to know

1. What did our classifier say about each email or post?
2. What should our classifier have said, i.e., the correct answer, usually as defined by humans ("gold label")

First step in evaluation: The confusion matrix

		<i>gold standard labels</i>	
		gold positive	gold negative
<i>system output labels</i>	system positive	true positive	false positive
	system negative	false negative	true negative

Accuracy on the confusion matrix

		<i>gold standard labels</i>	
		gold positive	gold negative
<i>system output labels</i>	system positive	true positive	false positive
	system negative	false negative	true negative

$$\text{accuracy} = \frac{\text{tp} + \text{tn}}{\text{tp} + \text{fp} + \text{tn} + \text{fn}}$$

Why don't we use accuracy?

Accuracy doesn't work well when we're dealing with uncommon or imbalanced classes

Suppose we look at 1,000,000 social media posts to find Delicious Pie-lovers (or haters)

- 100 of them talk about our pie
- 999,900 are posts about something unrelated

Imagine the following simple classifier

Every post is "not about pie"

Accuracy re: pie posts

100 posts are about pie; 999,900 aren't

gold standard labels

		gold positive	gold negative
<i>system output labels</i>	system positive	true positive	false positive
	system negative	false negative	true negative

$$\text{accuracy} = \frac{\text{tp} + \text{tn}}{\text{tp} + \text{fp} + \text{tn} + \text{fn}}$$

Why don't we use accuracy?

Accuracy of our "nothing is pie" classifier

999,900 true negatives and 100 false negatives

Accuracy is $999,900/1,000,000 = 99.99\%$!

But useless at finding pie-lovers (or haters)!!

Which was our goal!

Accuracy doesn't work well for unbalanced classes

Most tweets are not about pie!

Instead of accuracy we use precision and recall

		<i>gold standard labels</i>		
		gold positive	gold negative	
<i>system output labels</i>	system positive	true positive	false positive	precision = $\frac{tp}{tp+fp}$
	system negative	false negative	true negative	
		recall = $\frac{tp}{tp+fn}$		accuracy = $\frac{tp+tn}{tp+fp+tn+fn}$

Precision: % of selected items that are correct

Recall: % of correct items that are selected

Precision/Recall aren't fooled by the "just call everything negative" classifier!

Stupid classifier: Just say no: every tweet is "not about pie"

- 100 tweets talk about pie, 999,900 tweets don't
- Accuracy = $999,900 / 1,000,000 = 99.99\%$

But the Recall and Precision for this classifier are terrible:

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

A combined measure: F1

- F1 is a combination of precision and recall.

$$F_1 = \frac{2PR}{P + R}$$

F1 is a special case of the general "F-measure"

F-measure is the (weighted) harmonic mean of precision and recall

$$\text{HarmonicMean}(a_1, a_2, a_3, a_4, \dots, a_n) = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}}$$

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}} \quad \text{or} \left(\text{with } \beta^2 = \frac{1 - \alpha}{\alpha} \right) \quad F = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

F1 is a special case of F-measure with $\beta=1$, $\alpha=1/2$

Suppose we have more than 2 classes?

- Lots of text classification tasks have more than two classes.
 - Sentiment analysis (positive, negative, neutral) , named entities (person, location, organization)
- We can define precision and recall for multiple classes like this 3-way email task:

		<i>gold labels</i>			
		urgent	normal	spam	
<i>system output</i>	urgent	8	10	1	precision_u = $\frac{8}{8+10+1}$
	normal	5	60	50	precision_n = $\frac{60}{5+60+50}$
	spam	3	30	200	precision_s = $\frac{200}{3+30+200}$
		recall_u = $\frac{8}{8+5+3}$	recall_n = $\frac{60}{10+60+30}$	recall_s = $\frac{200}{1+50+200}$	

How to combine P/R values for different classes: Microaveraging vs Macroaveraging

Class 1: Urgent

	true urgent	true not
system urgent	8	11
system not	8	340

$$\text{precision} = \frac{8}{8+11} = .42$$

Class 2: Normal

	true normal	true not
system normal	60	55
system not	40	212

$$\text{precision} = \frac{60}{60+55} = .52$$

Class 3: Spam

	true spam	true not
system spam	200	33
system not	51	83

$$\text{precision} = \frac{200}{200+33} = .86$$

Pooled

	true yes	true no
system yes	268	99
system no	99	635

$$\text{microaverage precision} = \frac{268}{268+99} = .73$$

$$\text{macroaverage precision} = \frac{.42+.52+.86}{3} = .60$$

Text Classification and Naive Bayes

Precision, Recall, and F1

Text Classification and Naive Bayes

Avoiding Harms in Classification

Harms of classification

Classifiers, like any NLP algorithm, can cause harms

This is true for any classifier, whether Naive Bayes or other algorithms

Representational Harms

- Harms caused by a system that demeans a social group
 - Such as by perpetuating negative stereotypes about them.
- Kiritchenko and Mohammad 2018 study
 - Examined 200 **sentiment analysis** systems on pairs of sentences
 - **Identical** except for names:
 - common African American (Shaniqua) or European American (Stephanie).
 - Like "I talked to Shaniqua yesterday" vs "I talked to Stephanie yesterday"
- Result: systems assigned **lower sentiment** and more negative emotion to sentences with **African American names**
- Downstream harm:
 - Perpetuates stereotypes about African Americans
 - African Americans treated differently by NLP tools like sentiment (widely used in marketing research, mental health studies, etc.)

Harms of Censorship

- **Toxicity detection** is the text classification task of detecting hate speech, abuse, harassment, or other kinds of toxic language.
 - Widely used in online content moderation
- Toxicity classifiers incorrectly flag non-toxic sentences that simply mention minority identities (like the words "blind" or "gay")
 - women (Park et al., 2018),
 - disabled people (Hutchinson et al., 2020)
 - gay people (Dixon et al., 2018; Oliva et al., 2021)
- Downstream harms:
 - Censorship of speech by disabled people and other groups
 - Speech by these groups becomes less visible online
 - Writers might be nudged by these algorithms to avoid these words making people less likely to write about themselves or these groups.

Performance Disparities

1. Text classifiers perform worse on many **languages** of the world due to lack of data or labels
2. Text classifiers perform worse on **varieties** of even high-resource languages like English
 - Example task: **language identification**, a first step in NLP pipeline ("Is this post in English or not?")
 - English language detection performance worse for writers who are African American (Blodgett and O'Connor 2017) or from India (Jurgens et al., 2017)

Harms in text classification

- **Causes:**
 - Issues in the data; NLP systems amplify biases in training data
 - Problems in the labels
 - Problems in the algorithms (like what the model is trained to optimize)
- **Prevalence:** The same problems occur throughout NLP (including large language models)
- **Solutions:** There are no general mitigations or solutions
 - But harm mitigation is an active area of research
 - And there are standard benchmarks and tools that we can use for measuring some of the harms

Text Classification and Naive Bayes

Avoiding Harms in Classification

Let's do:
S06_1_Exercises

Let's review:
S06_2_Exercises_solution

Evaluation time!